

# SeeChat

## Internal Communication Platform

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### System Overview, Security & Review Guide

|                       |                                                 |
|-----------------------|-------------------------------------------------|
| CURRENT PHASE         | Internal Review                                 |
| DEPLOYMENT            | Internal Office Network                         |
| PROPOSED INTERNAL URL | <a href="https://seechat/">https://seechat/</a> |

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## 1. System Overview

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SeeChat is an internal communication application designed for employees working on the company network. It provides real-time text communication, group communication, user availability information and administrative controls within the internal environment.

|                               |                                                        |
|-------------------------------|--------------------------------------------------------|
| <b>System</b>                 | SeeChat                                                |
| <b>Purpose</b>                | Internal Team Communication                            |
| <b>Network</b>                | Internal Office Network                                |
| <b>Internet Requirement</b>   | Public internet is not required for core communication |
| <b>Current Authentication</b> | Local test authentication                              |
| <b>Planned Authentication</b> | Active Directory after approval                        |
| <b>Current Status</b>         | <b>Ready for Internal Review</b>                       |

## 2. Why SeeChat

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In daily office operations, communication challenges can arise:

- Employees may work inside full-screen applications and may miss browser or email communication.
- Teams may sometimes need quick internal communication.
- An internally managed communication option provides better control over access and availability.
- The IT team can manage user access centrally.
- The application is intended strictly for communication within the company network.

## 3. System Components

SeeChat is structured into three main parts:

### Frontend

The frontend is the interface employees use in the browser. Examples include: Team Directory, Chat Interface, Broadcast Groups, System Notifications, User Search, and Admin Console.

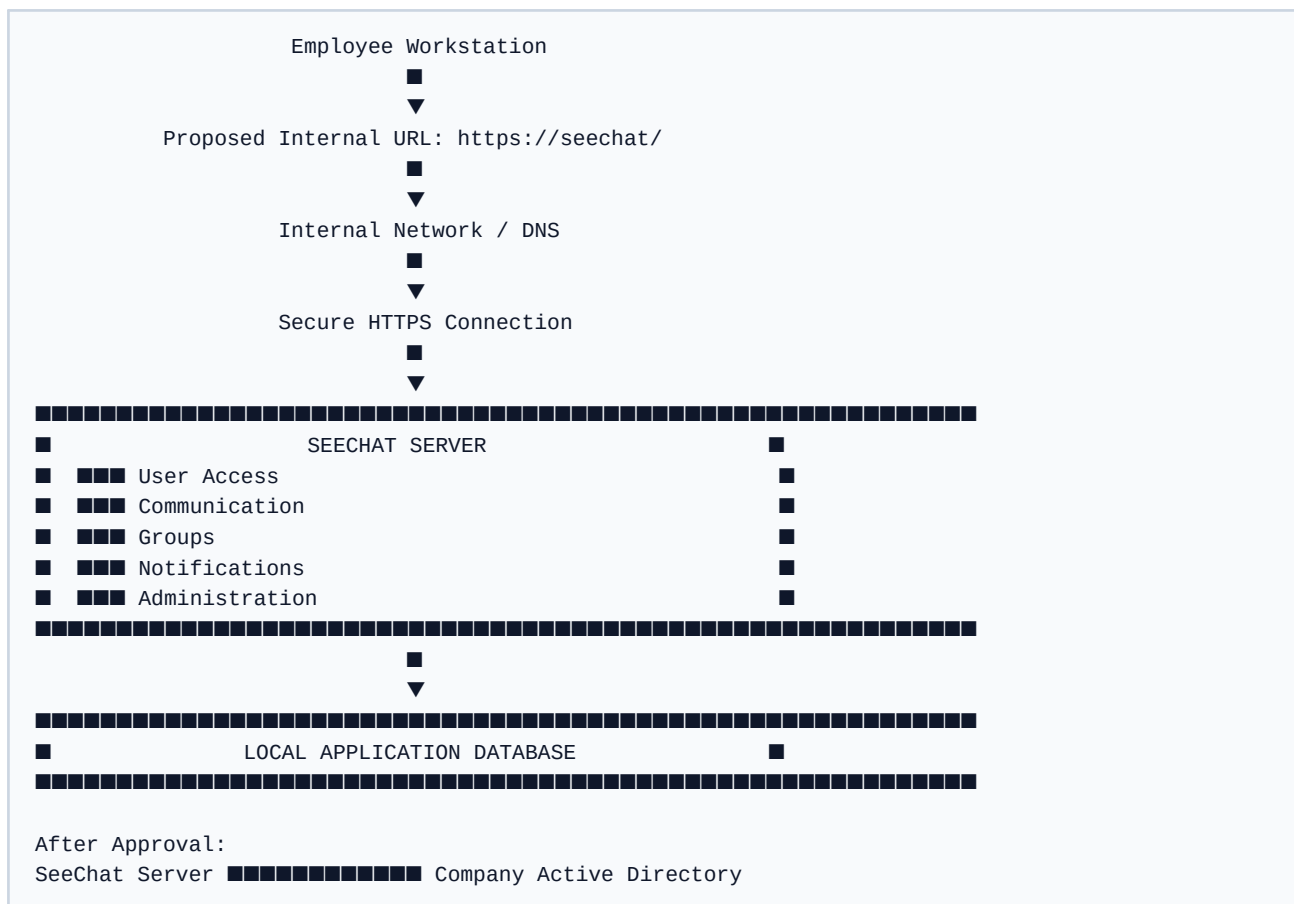
### Backend

The backend runs on the SeeChat server and manages application logic and communication. Functions include: validates user sessions, checks communication permissions, routes communication to intended recipients, manages broadcast groups, manages user status updates, and applies security rules.

### Database

The local application database maintains the application information required for user access, communication features, groups and status management.

## 4. System Flow



## 5. How SeeChat Works

|    |                                                                                                    |
|----|----------------------------------------------------------------------------------------------------|
| 01 | <b>Access Application:</b> User opens SeeChat at <a href="https://seechat/">https://seechat/</a> . |
| 02 | <b>Verify Access:</b> User access is verified by the server.                                       |
| 03 | <b>Display Directory:</b> Team Directory is displayed with user availability.                      |
| 04 | <b>Select Target:</b> User selects a team member or group.                                         |
| 05 | <b>Check Permission:</b> Communication permission is checked.                                      |
| 06 | <b>Send Message:</b> User sends a message.                                                         |
| 07 | <b>Validate Request:</b> SeeChat validates the request.                                            |
| 08 | <b>Route Communication:</b> Communication is routed to the intended user.                          |
| 09 | <b>Update Status:</b> Sent, Delivered and Read status is updated.                                  |
| 10 | <b>Update Unread State:</b> Unread indicators are updated automatically.                           |

## 6. Main Features

| CATEGORY           | CAPABILITIES & DESCRIPTION                                                                                                          |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Communication      | Direct Messaging, Chat Requests, Groups, Quoted Replies, Common Predefined Quick Replies.                                           |
| Message Controls   | Sent / Delivered / Read visual status, Unread Indicators, Typing Indicator, Emoji & Reactions, Edit/Delete Own Message, Clear Chat. |
| Workplace Features | Automatic Presence (Available / Away), Call to Desk urgent pings, Birthday Cards, Member Search, Conversation Pinning.              |
| Administration     | User Management, Enable / Disable Accounts, Role Management, Group Management, System Announcements, System Health.                 |

## 7. Security Controls

| CATEGORY               | SECURITY CONTROLS & PROTECTION                                                                                                                                                                   |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network                | Designed for internal network use; proposed HTTPS access; public internet not required for core communication.                                                                                   |
| User Access            | Authenticated user sessions; user identity checked by the server; admin actions restricted by role.                                                                                              |
| Application Protection | Input validation against malicious input; protection against common web input attacks; database queries use safe parameter handling; server-side rate limiting against excessive messaging.      |
| System Protection      | Does not disable antivirus or local security software; does not modify firewall automatically; does not create hidden startup persistence; TLS private keys are kept separate from the frontend. |

## 8. Current Verification Status

| Verification Metric           | Result    | Status |
|-------------------------------|-----------|--------|
| Automated Tests               | 37 / 37   | PASS   |
| Chrome                        | Verified  | PASS   |
| Edge                          | Verified  | PASS   |
| Server Startup                | Verified  | PASS   |
| Browser Console Errors        | 0         | PASS   |
| Server Errors                 | 0         | PASS   |
| Private Communication Routing | Verified  | PASS   |
| Offline Core Operation        | Verified  | PASS   |
| Code & Security Review        | Completed | PASS   |

## 9. Proposed HTTPS Setup

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The proposed internal address is: **https://seechat/**

The final hostname and HTTPS configuration will be confirmed with the IT team before deployment.

Technical support from the IT team may be required for:

- Internal hostname / DNS setup
- Trusted TLS certificate provisioning
- Workstation certificate trust
- Network or reverse proxy configuration if required
- Final hostname confirmation

*No public internet domain needs to be purchased if the required internal DNS and certificate infrastructure is available.*

## 10. Planned Active Directory Integration

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Active Directory integration is currently disabled and will be configured only after the required internal review and approval.

### Planned Access Flow:

User opens SeeChat ■■■■ User enters company credentials ■■■■ SeeChat validates account ■■■■  
Authorized user receives access

### Benefits:

- Users can use their existing company credentials for SeeChat access.
- Access can follow company account status.
- User account management can remain centralized with the IT team.

## 11. Review & Deployment Process

| PHASE     | STAGE DESCRIPTION                                                                                                   | STATUS        |
|-----------|---------------------------------------------------------------------------------------------------------------------|---------------|
| COMPLETED | Development, Internal Functional Testing & Code/Security Review                                                     | COMPLETED     |
| CURRENT   | Internal Management / IT / Security Review                                                                          | CURRENT STAGE |
| NEXT      | Endpoint / Antivirus Scan ■■■■ HTTPS/DNS Config ■■■■ AD Integration ■■■■ Controlled Testing ■■■■ Production Rollout | PENDING       |

## 12. System Boundaries

SeeChat is **not** designed to:

- Depend on public social media accounts.
- Depend on external cloud messaging for core communication.
- Provide unrestricted file sharing.
- Modify firewall settings automatically.
- Disable antivirus or security software.
- Create hidden system persistence.

## 13. Current Status

SeeChat has completed the current development and internal verification stage and is ready for management, IT and security review.

|                       |                                                                                                                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pending Review Items  | <ol style="list-style-type: none"><li>1. Final functional review</li><li>2. Endpoint / antivirus scan</li><li>3. IT and security review</li><li>4. HTTPS / DNS configuration confirmation</li><li>5. Management review / approval</li></ol> |
| Proposed Internal URL | <a href="https://seechat/">https://seechat/</a>                                                                                                                                                                                             |